

Task 5. Hitori

Available marks: 28

The Japanese Nikoli corporation produces an almost inexhaustable array of interesting logic puzzles. One of these is *Hitori*, dating from 1990.

Hitori is played on a square grid containing numbers in every cell. The aim of the game is to mark some of the numbers (for solutions in the paper game, marked cells are shown shaded) so that the remaining numbers are not duplicated in any row or column. Additional rules are

1. Only a minimum number of cells are to be marked (otherwise the game is trivial);
2. Marked cells must not be adjacent horizontally or vertically; and
3. Unmarked cells must form a single contiguous region, so that a path exists between any pair of cells by moving horizontally or vertically.

4	3	5	6	5	5
2	1	2	4	3	6
6	4	5	1	1	4
5	5	4	2	4	1
2	3	4	1	2	3
1	2	3	5	5	4

Format

Input and output formats are identical. Each file contains one or more games in some state. Each game starts with a line giving N , the order of the game (number of cells per row or column), followed by a short identifying name. Then follows the cell contents in row order, one row per line. Marked cells are represented by an asterisk following the value. Values lie between 1 and N . Spacing is such that the cell values align for readability whether or not they are marked. The last line contains a 0 and the name END.

The solved game pictured above could be represented by this input:

```
6 Hitori
4 3 5* 6 5* 5
2* 1 2 4 3 6
6 4 5 1* 1 4*
5* 5 4* 2 4 1
2 3* 4 1 2* 3
1 2 3 5* 5 4
0 END
```

Your Task

There are two parts to the task: validation, worth 12 marks, and solution, worth 16 marks.

For each game in an input file, report the name and the state it is in. If it violates rule 1, 2 or 3, report which rule it violates and one of the invalid cells. One error message is sufficient. (Rule 1 violations occur if a marked cell does not have an unmarked cell with the same value in either its row or column). If the game is already solved, report **Solved**. Otherwise report **Unsolved** and attempt to solve the game, displaying the solution. All valid inputs are solvable.

There are two test files, **task5A.dat** and **task5B.dat**, each with 4 test cases. The maximum order of any puzzle is 9.

Note: If a game in **task5B.dat** takes more than 5 seconds to solve it counts as a failed test.

Continued over...

Sample Data and Output

Input (**task5sample.dat**):

```
4 Sample1
2* 2 2* 4
1 4 2 3
2 3 2* 1
3 4* 1 2
5 Sample2
3 5 4 1 2
4* 3 2* 2 3*
4 3* 2 5 1
1 2* 5 3 4
2* 2 3 2* 5
5 Sample3
4 5 3 4* 1
2* 1 4* 5 2
3 3* 5 1 3*
1 3 3* 2 5
3* 4 2 3 1*
0 END
```

Output:

```
Sample1 is Solved
Sample2 violates rule 2: cell 3* at [2,1] (0-origin) has an adjacent marked cell
Sample3 violates rule 1: cell 4* at [1,2] (0-origin) has no matching value in its row or column
```

References:

[Conceptis Puzzles](#)

Vegard Hanssen's [database of Hitori puzzles](#)